

= 0.5. Since it is a case of coin tosses with two outcomes (H and T), the binomial probability distribution is applicable. As you know, in the case of binomial distribution, the variance  $\sigma^2$  is  $npq$ . Thus,  $\sigma = \sqrt{npq}$ . Hence, the term with  $\sqrt{n}$  in the denominator cancels out. The central limit theorem (CLT) as per equation (2.4) tells us that

$$Z = \frac{\bar{X} - 0.5}{\sqrt{pq}} \sim N(0, 1)$$

$$Z = \frac{0.43 - 0.5}{\sqrt{0.5(1-0.5)}} = \frac{-0.07}{0.5} = -0.14$$

The critical value of  $Z$  at 5 per cent level of significance is 1.96. The Calculated value of  $Z$  (absolute) is less the critical  $Z$  value. We therefore accept the null hypothesis that the coin is not biased. We conclude that the data from the trials does not suggest that the coin is biased.

### Check Your Progress 1

- 1) Differentiate between a parameter and a statistic.

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- 2) Distinguish between a 'point estimate' and an 'interval estimate'.

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- 3) What is meant by the term 'sampling distribution of a statistic'?

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- 4) State the properties of a 'good estimator'.

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- 5) What are the two conditions that a consistent estimator should satisfy?

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- 6) Differentiate between the weak and strong versions of the law of large numbers.

## 2.4 HYPOTHESIS TESTING

A statistical hypothesis is a statement about a population or equivalent about a population parameter. Such a statement made may or may not be true. Hence, we wish to test the hypothesis on the basis of the evidence from a random sample. If the hypothesis completely specifies the population, then it is known as a 'simple hypothesis'; else, it is known as a 'composite hypothesis'. For instance, in a sampling from a normal population  $N(\mu, \sigma^2)$ , the hypothesis  $H: \mu = \mu_0, \sigma^2 = \sigma_0^2$  is a simple hypothesis. It is complete because it specifies a value for both the parameters of the distribution. On the other hand, each of the following hypothesis is composite hypothesis.

- i)  $\mu = \mu_0$  (nothing is specified about  $\sigma^2$ )
- ii)  $\sigma^2 = \sigma_0^2$  (nothing is specified about  $\mu$ )
- iii)  $\mu < \mu_0, \sigma^2 = \sigma_0^2$  ( $\mu$  is not exactly specified).

### 2.4.1 Null Hypothesis and Alternative Hypothesis

The null hypothesis is an assertion whose validity is to be tested for possible rejection based on sample observations. It is denoted by  $H_0$ . It is tested against other options specified in the 'alternative hypothesis'. An alternative hypothesis is indicated by  $H_1$ . The alternative hypothesis is not tested, but its acceptance or rejection is determined by the null hypothesis's rejection or acceptance. In a strict sense, we actually either 'reject' or 'do not reject' a null hypothesis. Though the latter (i.e., do not reject) is commonly referred to as 'acceptance', it is actually the case of 'not rejecting the null hypothesis'. Therefore, you should always remember that a hypothesis is never accepted; it is either rejected or not rejected.

The null hypothesis that the population mean  $\mu$  takes a specific value  $\mu_0$  is expressed as  $H_0: \mu = \mu_0$ . Now, the alternative hypothesis may be expressed as:  $H_1: \mu \neq \mu_0$  or  $H_1: \mu > \mu_0$  or  $H_1: \mu < \mu_0$ . This brings us to the concepts of 'one-tailed test' and 'two-tailed test'. A null hypothesis tested against the alternative hypothesis like  $\mu < \mu_0$  or  $\mu > \mu_0$  is called a one-tailed test. More specifically, if the alternative hypothesis is  $H_1: \mu > \mu_0$  it is called a right tailed test. Likewise, if the alternative hypothesis is  $H_1: \mu < \mu_0$  it is called a left-tailed. Both these are cases of 'one-tailed test'. A null hypothesis where the alternative hypothesis is  $H_1: \mu \neq \mu_0$  is known as 'two-tailed test'.

### 2.4.2 Type-I Error and Type-II Error

The error committed in rejecting a null hypothesis by the test, when it is actually true, is termed as Type-I error. The error committed in accepting a null

hypothesis by the test, when it is actually false, is termed as Type-II error. The correct decision by the test procedure followed, and the likely error, is depicted in Table 2.1.

**Table 2.1: Error in Hypothesis Testing**

|                | Ho is True      | Ho is False      |
|----------------|-----------------|------------------|
| Ho is accepted | ✓               | Error of Type-II |
| Ho is rejected | Error of Type-I | ✓                |

### 2.4.3 Level of Significance

The level of significance refers to the maximum permitted level of committing Type-I error. It is a level of risk that we are willing to take. It is denoted by  $\alpha$  and indicates the probability of rejecting the null hypothesis [i.e.,  $P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true}) = \alpha$ ]. The most commonly used levels of significance in practice are 5 per cent and 1 per cent. If we use a 5 per cent level of significance, this means that we are likely to reject a 'right  $H_0$ ' in 5 out of 100 samples. To put it another way, this means we have a 95 per cent confidence level of not committing the Type-I error. Thus, at  $\alpha$  level of significance, the degree of confidence in our decision is  $(1-\alpha)$  which is also called the 'confidence coefficient'. The  $p$ -value is thus defined as the lowest significance level at which a null hypothesis can be rejected.

### 2.4.4 Critical Region

The critical region indicates the probability that the Type-I error committed in our statistical inference does not exceed the level of significance. Let us consider the drawing of numerous samples of the same size from a population and compute the same statistic  $t$  (say  $\bar{x}$ ) for each one of them. Let  $t_1, t_2, \dots, t_k$  be the sample value of the statistic for these samples. Now, each one of these values can be used to test the null hypothesis  $H_0$ . Some values might cause  $H_0$  to be rejected, while others might cause  $H_0$  to be accepted. This sample statistic  $t_1, t_2, \dots, t_k$  can therefore be separated into two mutually exclusive regions. While one leads to  $H_0$ 's rejection, the other leads to  $H_0$ 's acceptance. The region that leads to  $H_0$ 's rejection is referred to as the 'critical region', while that which leads to  $H_0$ 's acceptance is referred to as the 'acceptance region'. The test statistic's critical value is the value for which the test simply rejects the null hypothesis at the given significance level.

### 2.4.5 Confidence Interval

Due to random sampling error, it is impossible to know the exact value of the population parameter using only the information in the sample. However, we can generate a collection of values (or set of values) containing the population

parameter with a pre-specified probability using the data from a random sample. Such a pre-specified probability, which says that the population parameter is contained in this set, is termed as the 'confidence level'. Such a set is called a 'confidence set'. The population parameter's confidence set represents all possible values of the population parameter lying between the lower limit and the upper limit. This is called the 'confidence interval'.

## 2.5 ESTIMATION METHODS

There are three methods of estimating the parameters of a population regression function. These are: (i) method of least squares, (ii) method of maximum likelihood and (iii) method of moments.

### 2.5.1 Method of Least Squares

Consider a system of  $m$  linear equations with  $n$  unknowns as follows:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \dots & \dots \dots \\ \dots & \dots \dots \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned} \quad \dots(2.5)$$

where,  $a_{ij}$  ( $i = 1, 2, \dots, m; j = 1, 2, \dots, n$ ) and  $b_i$  ( $i = 1, 2, \dots, m$ ) are constants and  $x_j$  ( $j = 1, 2, \dots, n$ ) are the ' $n$ ' unknowns. The system of equations (2.5) will not have a unique solution if the number of equations is more than the number of unknowns ( $m > n$ ). In such a case, we have to find a set of values of  $x_j$  ( $j = 1, 2, \dots, n$ ) which best satisfies the equations in (2.5). This means, such a set of solution should be as close as possible to the actual values of the unknowns. This is achieved by a method known as the 'method of least squares'.

Note that for the equations (2.5) [which are like  $y_i = a + bx_i + e_i$ ], for a set of values  $x_1, x_2, \dots, x_n$ ,  $e_i = a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n - b_i$ ,  $i = 1, 2, \dots, m$  are the residues. Hence,  $S = \sum_{i=1}^m e_i^2$  is the 'sum of the squares of the residues'. The least squares principle states that the best values of the unknown are those for which the 'sum of the squares of the residues' is the least. Hence, to compute the values of  $x_j$  ( $j = 1, 2, \dots, n$ ) such that  $S$  is minimum, we need to have  $dS/dx_j = 0, j = 1, 2, \dots, n$ . That is,

$$dS/dx_1 = dS/dx_2 = \dots = dS/dx_n = 0 \quad \dots(2.6)$$

The conditions in (2.6) gives  $n$  equations involving  $n$  unknowns. These equations are called normal equations. Solving these  $n$  normal equations with  $n$  unknowns we get a unique best solution. Let us take an example to understand this better. Say we want to fit a straight line  $y = a + bx$  for a set of a given points  $(x_i, y_i)$ ,  $i = 1, 2, \dots, n$ . For any  $x_i$  the estimated value is  $\hat{y}_i = a + bx_i$ . Now,  $S$ , the Sum of the squares of the residuals is:

$$S = \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \sum_{i=1}^n (a + b x_i - y_i)^2$$

Applying the method of least squares to calculate  $a$  and  $b$ , requires us to have:

$$dS/da = 0, dS/db = 0, \text{ and } d^2S/da^2 > 0, d^2S/db^2 > 0, \quad \dots (2.7)$$

The conditions (2.7) give two normal equations as follows:

$$\sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i$$

and

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad \dots (2.8)$$

Solving the two equations in (2.8), we can find the values of  $a$  and  $b$ . If  $\mathbf{a}^*$  and  $\mathbf{b}^*$  are the calculated values of  $a$  and  $b$  respectively, then the best fitted line will be

$$y = \mathbf{a}^* + \mathbf{b}^* x$$

### 2.5.2 Method of Maximum Likelihood

Let  $X_1, X_2, \dots, X_n$  be  $n$  random variables from a population with the probability density function (p.d.f.)  $f(X, \theta)$  where  $\theta$  is the unknown parameter. The probability is a function of sample values  $X_1, X_2, \dots, X_n$  and the parameters  $\theta_1, \theta_2, \dots, \theta_k$ . The likelihood function is expressed as:

$$L(\theta) = L(X_1, X_2, \dots, X_n, \theta_1, \theta_2, \dots, \theta_k)$$

The maximum likelihood estimator (MLE) is derived by using the principal of maximum and minimum to the above likelihood function as:

$$\frac{dL}{d\theta} = 0 \text{ and } \frac{d^2L}{d\theta^2} < 0 \quad \dots (2.9)$$

Since  $L > 0$ ,  $\log L(\theta)$  attains its maximum for the same value of  $\theta$  as  $L(\theta)$ . Thus, the maximum likelihood estimator of  $\theta$  is derived by differentiating the  $\log L(\theta)$ , i.e., the value of  $\theta$  for which:

$$\frac{d \log(L(\theta))}{d\theta} = 0 \text{ and } \frac{d^2 \log(L(\theta))}{d\theta^2} < 0$$

First let us consider the case where all  $x_i$ 's are discrete. This means,  $P[\theta, (X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)]$  is a function of  $\theta$  and  $x_1, x_2, \dots, x_n$ . For fixed  $(x_1, x_2, \dots, x_n)$ , we therefore have:

$$\begin{aligned} L(\theta) &= P[\theta, (X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)] \\ &= P[\theta, X_1 = x_1] \cdot P[\theta, X_2 = x_2] \cdot \dots \cdot P[\theta, X_n = x_n] \cdot [\text{since } x_1, \dots, x_n \text{ are independent}] \\ &= \prod_{i=1}^n P[\theta(X_i = x_i)] = \prod_{i=1}^n f[\theta(x_i)] \end{aligned}$$

where  $f$  is the common 'probability function' (either a mass function for discrete variables or a density function for continuous variables). A value  $\hat{\theta}$  of  $\theta$  is now the maximum likelihood estimator, if  $L(\hat{\theta}) \geq L(\theta)$  for all  $\theta$ . In most cases, we take  $\log L(\theta)$ , so that further treatments become easier since  $\log$  is a one-to-one function. The equation,  $\frac{d(\log L(\theta))}{dL(\theta)} = 0$ , is known as the likelihood equation.

We can similarly consider the case where all  $x_i$ 's are continuous where  $L(\theta)$  is the joint  $p.d.f$  of  $(x_1, x_2, \dots, x_n)$ ,  $dx_1, dx_2, \dots, dx_n$

$$\begin{aligned}\therefore L(\theta) &= f(x_1, x_2, \dots, x_n), dx_1, dx_2, \dots, dx_n \\ &= \prod_{i=1}^n f(x_i) \text{ (since } X_i \text{'s are independent)}\end{aligned}$$

A likelihood function is thus the product of individual probability functions. Let us now consider the MLEs of some standard distributions.

**Binomial Distribution:** Let  $X_i$  be  $B(m, p)$  with pmf given by  $f_p(x_i) = mC_{x_i} p^{x_i} (1-p)^{m-x_i}$ , i.e., each  $x_i$  is an independent binomial variable with  $(m, p)$  as its parameters. Therefore, the likelihood function of  $x_i$  is given by:

$$\begin{aligned}L(x_i) &= \prod_{i=1}^n f_p(x_i) \\ &= \prod_{i=1}^n mC_{x_i} p^{x_i} (1-p)^{m-x_i} \\ &= (\prod_{i=1}^n mC_{x_i}) p^{\sum x_i} (1-p)^{nm - \sum_{i=1}^n x_i}\end{aligned}$$

Taking log on both sides, we get:

$$\text{Log } L(x_i) = \text{Log} (\prod_{i=1}^n mC_{x_i}) + \sum x_i \log p + (nm - \sum x_i) \log(1-p) \quad \dots (2.10)$$

Differentiating equation (2.10) with respect to  $p$  we get

$$\begin{aligned}\frac{d \log L(x_i)}{dp} &= \frac{\sum x_i}{p} + \frac{nm - \sum x_i}{(1-p)} \cdot (-1) \\ \frac{L'(x_i)}{L(x_i)} &= \frac{\sum x_i}{p} - \frac{nm - \sum x_i}{(1-p)} \\ &= \frac{\sum x_i - p \sum x_i - mnp + p \sum x_i}{p(1-p)} \\ &= \frac{\sum x_i - mnp}{p(1-p)}\end{aligned}$$

Equating the first order condition to zero we get

$$\frac{\sum x_i - mnp}{p(1-p)} = 0$$

$$\text{or } p = \frac{\sum x_i}{mn} = \hat{p}$$

Therefore,  $\hat{p} = \frac{\sum x_i}{mn}$  is the maximum likelihood estimator of  $p$ .

**Poisson Distribution:** Now, let  $x_1, x_2, \dots, x_n$  be a sample of  $n$  observations drawn from a Poisson distribution having parameter  $m$ . The probability mass function (pmf) of  $x$  is given by:

$$f_m(x) = \frac{e^{-m} \cdot m^x}{x!} \quad (x = 0, 1, 2, \dots, \infty)$$

$$\begin{aligned}\therefore L(x) &= \prod_{i=1}^n f_m(x_i) \\ &= \prod_{i=1}^n \frac{e^{-m} \cdot m^{x_i}}{x_i!}\end{aligned}$$

Taking log on both sides, we get:

$$\begin{aligned}
\text{Log } L(x) &= \log(\prod_{i=1}^n e^{-m}) + \sum x_i \log m - \sum_{i=1}^n \log x_i! \\
&= \log e^{-nm} + \sum x_i \log m - \sum_{i=1}^n \log x_i! \quad \dots (2.11)
\end{aligned}$$

Differentiating equation (2.11) with respect to  $m$  we get,

$$\frac{\partial \text{Log } L(x)}{\partial m} = -n + \frac{\sum x_i}{m}$$

Equating the first order condition (FOC) to zero we get:

$$-n + \frac{\sum x_i}{m} = 0$$

$$\frac{\sum x_i}{m} = n$$

$$\frac{\sum x_i}{n} = m$$

$$\bar{x} = m = \hat{m}$$

$\therefore \hat{m} = \bar{x}$  is the MLE of the Poisson parameter  $m$ .

Normal Distribution: Let us now derive the MLE of the parameter  $\sigma^2$  in a normal distribution with  $N(\mu, \sigma^2)$ . Let us first assume that the parameter  $\mu$  is known. Now, the probability density function (pdf) of the normal variable  $X$  is:

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} - \infty < x < \infty$$

The likelihood function is given by

$$\begin{aligned}
L(x; \mu, \sigma^2) &= -n \log \sigma - \frac{n}{2} \log(2\pi) - \frac{\sum (x_i - \mu)^2}{2\sigma^2} \\
&= -\frac{n}{2} \log \sigma^2 - \frac{n}{2} \log(2\pi) - \frac{\sum (x_i - \mu)^2}{2\sigma^2} \quad \dots (2.12)
\end{aligned}$$

Differentiating equation (2.12) with respect to  $\sigma^2$  we get,

$$\frac{\partial \text{Log } L(x; \mu, \sigma^2)}{\partial \sigma^2} = \frac{-n}{2\sigma^2} + \frac{\sum (x_i - \mu)^2}{2\sigma^4}$$

$\therefore$  Applying the FOC = 0 we obtain

$$\frac{-n}{2\sigma^2} + \frac{\sum (x_i - \mu)^2}{2\sigma^4} = 0$$

$$\text{or, } \sigma^2 = \frac{\sum (x_i - \mu)^2}{n} = \hat{\sigma}^2$$

$\therefore \hat{\sigma}^2$  is the M. L. E. of  $\sigma^2$

Note that  $\hat{\sigma}^2$  is an unbiased estimator of  $\sigma^2$  because:

$$\begin{aligned}
E(\hat{\sigma}^2) &= E\left[\frac{\sum (x_i - \mu)^2}{n}\right] \\
&= \frac{1}{n} E[\sum (x_i - \mu)^2]
\end{aligned}$$

$$= \frac{1}{n} \cdot n \sigma^2 = \sigma^2$$

Let us now consider the case where both  $\mu$  &  $\sigma^2$  are unknown. The likelihood function is given by:

$$\text{Log } L = -\frac{n}{2} \text{Log } \sigma^2 - \frac{n}{2} \text{Log}(2\pi) - \frac{\sum (x_i - \mu)^2}{2\sigma^2} \quad \dots (2.13)$$

Differentiating equation (2.13) with respect to  $\mu$  we get:

$$\begin{aligned} \frac{d \text{Log } L}{d\mu} &= -\frac{2\sum (x_i - \mu)}{2\sigma^2} (-1) \\ &= -\frac{\sum (x_i - \mu)}{\sigma^2} \end{aligned}$$

Applying FOC = 0, we get:  $\frac{\sum (x_i - \mu)}{\sigma^2} = 0$  or  $\sum x_i = n\mu$

$$\therefore \frac{\sum x_i}{n} = \mu \text{ or } \bar{x} = \mu = \hat{\mu}$$

$\bar{x}$  is the MLE of  $\mu$ . Likewise:

$$\hat{\sigma}^2 = \frac{\sum (x_i - \mu)^2}{n} = s^2 \text{ [sample variance] is the MLE of } \sigma^2.$$

Properties of MLE: Based on the above discussion, we can list the properties of MLE as follows:

- i) The MLE is consistent, efficient and sufficient.
- ii) The MLE is not necessarily unbiased. But when the MLE is biased, by a slight modification, it can be converted in to an unbiased estimator.
- iii) The MLE tends to be distributed normally for large sample.
- iv) The MLE is invariant under functional transformation. This means if  $T$  is an MLE of  $\theta$ , and  $g(\theta)$  is a function of  $\theta$ , then  $g(T)$  is an MLE of  $g(\theta)$ .

### 2.5.3 Method of Moments

When we fit a particular observed distribution based on a sample of observations, our objective is to see whether the observed distribution is a random sample from its parent distribution. Therefore, before estimating the parameters of a theoretical distribution from the observed data, the first step is the fitting of a distribution. This is particularly the case when the parameter values are assumed or known. In such cases, the method of moments is an alternative method for estimating the parameters of a parent or theoretical distribution. In this method, the first  $K$  moments of the theoretical distribution expressed in terms of parameters are equated to their corresponding sample moments. In the second step, the  $K$  equations are solved to obtain the required population estimates.

Let  $X_1, X_2, \dots, X_n$  be a random sample of size  $n$  drawn from a population. The  $k^{\text{th}}$  order raw moment is given by:  $m_k = \frac{\sum_{i=1}^n X_i^k}{n}$ . Therefore,  $E(m_k) = \mu_k = 1/n \cdot E(\sum X_i^k)$ . You should note that  $\mu_k$  will be a function of the underlying parameters in most cases. We produce  $K$  equations by solving which we can get



estimates of  $K$  unknown parameters. The moments based on powers of  $X$  provide a source of parameter information. Let us now see some application of this method of estimation to some standard distributions.

Let  $X_1, X_2, \dots, X_n$  be Bernoulli random variables with parameter  $p$ . What is the moment's estimator of  $p$ ? For Bernoulli distribution, the first moment about zero is:  $E(X_i) = p$ . Hence, we have to derive the method of moments estimator for the only parameter  $p$  for which only one equation is required. Hence, equating the first theoretical moment about zero with the corresponding sample moment, we get:  $p = \frac{1}{n} \sum_{i=1}^n x_i$ . Therefore, the moment estimator for  $p$  is:  $\hat{p} = \frac{1}{n} \sum_{i=1}^n x_i$ .

Now, consider a sample of drawn from  $n$  i.i.d. random variables,  $X_1, X_2, \dots, X_n$ , with mean  $\mu$  and variance  $\sigma^2$ . What are the moments estimates of  $\mu$  and  $\sigma^2$ ? We evidently need two moments now. The first moment about zero of a normal distribution is  $E(X_i) = \mu$ . The second moment about zero of a normal distribution is  $E(X_i^2) = \sigma^2 + \mu^2$ . We thus have the two equations required to derive the estimates of two parameters. Equating the first theoretical moment about zero with the corresponding sample moment, we get:

$$E(X_i) = \mu = \frac{1}{n} \sum_{i=1}^n X_i \quad \dots (2.14)$$

Equating the second theoretical moment about zero with the corresponding sample moment, we get:

$$E(X_i^2) = \sigma^2 + \mu^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 \quad \dots (2.15)$$

Equation (2.14) gives the moments estimator of mean  $\mu$ , the sample mean, as:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

Likewise, from equation (2.15), we get:

$$\begin{aligned} \sigma^2 &= \frac{1}{n} \sum X_i^2 - \mu^2 \\ &= \frac{1}{n} \sum X_i^2 - \bar{x}^2 \\ &= \frac{1}{n} \sum (X_i - \bar{x})^2 \end{aligned}$$

Hence, the method of moments estimator of variance  $\sigma^2$  is given by

$$\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{x})^2$$

### Check Your Progress 2

- 1) When is a hypothesis said to be 'complete'? By what alternative name is it called?

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- 2) Distinguish between a 'one-tailed test' and a 'two-tailed test'.

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- 3) Distinguish between 'type-I error' and 'type-II error'. How is it related to the 'level of significance'?

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- 4) State the principle behind the 'method of least squares'. When will it have a 'unique solution'?

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- 5) State the properties of a maximum likelihood estimator.

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## 2.6 LET US SUM UP

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In this unit we discussed the classical statistical inference theory. Classical statistical inference consists of two parts: estimation and hypothesis testing. Estimation is the process of estimating a population parameter using the data drawn as a sample from the population. Hypothesis testing is the process of evaluating whether to accept or reject a hypothesis or assumption made about the value of a population parameter based on sample data. Under estimation, the unit has explained the concepts of point estimation and interval estimation. Properties of a good estimator are also discussed. Different estimation methods such as Method of Least Squares, Method of Maximum Likelihood and Method of Moments are discussed with illustrations.

## 2.7 KEY WORDS

- Point Estimation** : Specific value of a ‘statistic’ where the term statistic means a function of sample observations (e.g., mean, standard deviation).
- Interval Estimation** : It refers to the range within which the estimated value of the parameter lies. Thus it refers to a specific interval which depends upon the standard error and the level of significance. If  $T_1$  and  $T_2$  are two functions based on sample observations such that  $P(T_1 \leq \theta \leq T_2) = (1 - \alpha)$ , (where  $\alpha$  is the level of significance), then the interval  $(T_1, T_2)$  is termed as an ‘interval estimate’ or a confidence interval.
- Independent and Identically Distributed (i.i.d.)** : A collection of random variables is said to be independent and identically distributed if they fulfil two conditions: (i) each random variable has the same distribution (say, for example, normal or Poisson), and (ii) all the random variables are independent (it implies that knowledge of the value of one variable does not provide any clue about the value of the other variables). In symbols,  $P(AB) = P(A) \times P(B)$ .
- Law of Large Numbers (LLN)** : Tells us how sample mean can be considered a true representative estimate of population mean with the sample size getting infinitely large. There are two versions of this law, viz., strong LLN and weak LLN. The SLLN says that if  $A_n$  is the sample mean and  $\mu$  is the population mean, then  $\lim_{n \rightarrow \infty} P(|A_n - \mu| = 0) = 1$ . The WLLN, on the other hand, tells us that for any  $\alpha > 0$ ,  $\lim_{n \rightarrow \infty} P(|A_n - \mu| < \alpha) = 1$  or  $\lim_{n \rightarrow \infty} P(|A_n - \mu| > \alpha) = 0$ .
- Central Limit Theorem** : This result says that if  $X_i, i = 1, 2, \dots, n$  are i.i.d.  $N(\mu, \sigma^2)$ , then, as  $n \rightarrow \infty$ ,  $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ . Further, the standardised normal variate,  $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$  is  $N(0, 1)$ .
- Method of Least Squares** : The least squares principle states that the best values of the unknown are those for which the ‘sum of the squares of the residues’ is the least.
- Likelihood Function** : If  $f(X, \theta)$  is the probability function, then its likelihood function is defined as:  $L(\theta) = \prod_{i=1}^n f(\theta(x_i))$ .